

CHRISTOPHER LAM

clbears@student.ubc.ca · (925) 519-7632 · Pleasanton, CA · clbears.com

EDUCATION

University of British Columbia — Vancouver, Canada

Sept 2025 – Present

B.A.Sc., Engineering (Mechanical/Electrical)

Key Coursework: PHYS 159 (Experimental Physics Lab), PHYS 158 (Electricity, Magnetism & Optics), PHYS 157 (Thermodynamics, Oscillations & Waves), APSC 160 (Engineering Computation), APSC 100 (Engineering Design)

Foothill High School — Pleasanton, CA

Aug 2021 – May 2025

Diploma, GPA: 4.25

Key Coursework: AP Physics C: Mechanics & E&M, PLTW Digital Electronics, PLTW Computer Integrated Manufacturing (CAD/CAM, CNC), PLTW Principles of Engineering

ENGINEERING PROJECTS

Custom LED Desk Lamp

Independent Project

- Designed and built a desk lamp combining a spring-balanced mechanical arm with a custom analog LED driver circuit (3000K COB); tuned spring geometry for balance across positions and selected analog dimming for flicker-free operation at any brightness level

CNC Press-Fit Assembly — PLTW Computer Integrated Manufacturing

Foothill High School

- Machined a two-part press-fit assembly on a Tormach CNC mill — a thin-wall hex extrusion and mating blind pocket; iterated prototype feeds/speeds to prevent chatter, then dialed tolerance to audible separation on a single-shot final part

COVID-19 Emergency PPE Production

Pleasanton, CA, 2020

- Manufactured and distributed 3,500+ 3D-printed face shields to first-responder organizations; iterated on production workflow to maximize throughput; recognized by NorCal Ambulance

Mousetrap-Powered Vehicle — 1st Place

Foothill High School

- Engineered an original design using lightweight metal wire frame (50% weight reduction), rubber-band torque normalization, and balloon-friction wheels; outperformed entire junior class cohort

3D Printer Build & Fabrication

Independent Project

- Assembled and calibrated a large-format FDM printer from components, dialing in stepper, bed leveling, and extrusion parameters; used as a dedicated fabrication platform for CAD-modeled functional parts and prototypes

For a more thorough list, please see clbears.com

LABORATORY EXPERIENCE

PHYS 159 – Experimental Physics Laboratory

UBC, Jan 2026 – Present

- Designed experiments using CERT methodology for standing wave analysis; performed uncertainty propagation, oscilloscope measurements, and systematic data collection

LEADERSHIP & ACTIVITIES

Scouts BSA Troop 941 — Senior Patrol Leader & Eagle Scout

Pleasanton, CA

- Elected leader of a 100-member troop; directed meetings, outings, and leadership training — Eagle Scout Project: designed and led installation of a public bicycle repair station, coordinating 15+ volunteers
- Northern Tier High Adventure: Led crew on a 50-mile canoe expedition; navigated a thunderstorm crisis requiring real-time decision-making under wilderness constraints

Student Filmmakers Association — President

Foothill High School, 2024 – 2025

- Led weekly club meetings; coordinated short film production, cinematography workshops, script development, and merchandise design

WORK EXPERIENCE

In-N-Out Burger — Store Associate

Pleasanton, CA, 2022 – 2025

- Delivered high-quality service in a fast-paced environment; maintained precision and consistency standards under sustained pressure

SKILLS

Engineering Tools: Autodesk Inventor, Fusion 360, Arduino, CAD/CAM, 3D printing (FDM), CNC machining, laser cutting, soldering, oscilloscope

Digital Electronics: Circuit logic design, Boolean algebra, K-maps, breadboarding, Multisim simulation, troubleshooting

Programming: Python, Arduino C

Lab & Research: Uncertainty propagation, CERT experimental design, measurement systems, materials testing, prototyping

Languages: Japanese (conversational, 3 years)

Interests: Hiking (Mt. Whitney, Mt. Fuji), photography, design, film